# Customer Mobile App

## Detailed Delivery Roadmap

PROJECT

Native customer shopping app · Android + iOS

STATUS

● All 4 phases complete · Type-safe · Lint-clean

STACK

Expo SDK 54 · React Native 0.81 · NativeWind v4

ARCHITECTURE

Single Expo codebase · Same monorepo as web · Shared backend

DOCUMENT VERSION

v1.0 · May 2026

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*One backend. Two storefronts. Zero drift.*

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# 1 · Vision & Strategy

## What we set out to build

A **native customer storefront** for Android and iOS, designed in the spirit of Blinkit-style speed-first commerce while staying on-brand with Xelnova's mint-green identity. The app reaches **full feature parity with the existing Xelnova web storefront** and adds capabilities only possible on a real phone: push notifications, biometric-grade secure token storage, and a tap-driven shopping flow that feels at home next to Amazon, Flipkart and Blinkit on the user's home screen.

## Why mobile (and why now)

### CUSTOMER REACH

Over 70 % of Indian e-commerce traffic now originates on mobile. A web app loaded inside a phone browser leaves discoverability, retention and engagement on the table — push notifications alone lift order frequency by 25 – 40 % in benchmarks.

### OPERATIONAL LOCK-IN

Once a customer installs an app, the cost of switching to a competitor is substantially higher than re-typing a URL. Loyalty programs, saved cards, addresses, and notification permissions all reinforce this.

### STRATEGIC OPTION VALUE

The same monorepo and component library that powers this customer app can power a *seller* mobile app and a *business* mobile app at a fraction of the cost — between 60 % and 70 % of the foundational work is already done.

### BRAND INTEGRITY

Showing up in the App Store and Play Store with a polished native app signals scale and seriousness that a mobile website cannot.

## Design direction

The visual language is a **Blinkit-leaning hybrid**. Concrete decisions:

- **Dense product surfaces** — three-up grids, larger product images, prominent price chips
- **One-tap actions** — "Add to cart" and "Buy now" never more than two taps from any product
- **Sticky context** — location chip, search bar, cart badge always visible at the top of Home and Search
- **Brand fidelity** — mint primary (`#11ab3a`), deep mint (`#0c831f`), pale mint surfaces — extracted from the web theme so identity stays consistent across channels
- **Accessibility built in** — generous `hitSlop` on every interactive element, semantic text scales, sufficient contrast ratios, all icons paired with text labels

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# 2 · System Architecture

## Monorepo layout

```
xelnova-web-new (root)
├── apps/
│   ├── web              ← existing customer web app
│   ├── admin            ← admin panel (web)
│   ├── seller           ← seller dashboard (web)
│   ├── business         ← B2B portal (web)
│   ├── api              ← Next-side API routes
│   └── mobile-web       ← NEW: customer mobile app
│       ├── app/         ← Expo Router file-based screens (40+)
│       ├── src/         ← lib, components, hooks
│       └── docs/        ← client-facing artifacts (this PDF)
├── backend/             ← NestJS API service (unchanged)
└── packages/
    ├── api              ← shared HTTP client (unchanged)
    ├── utils            ← shared utilities (unchanged)
    └── ui-native        ← NEW: shared React Native components
```

## Why this layout works for the business

| | |
|---|---|
| **Single source of truth** | One backend powers web *and* mobile. Every endpoint is contractually identical, so a coupon launched on web works on mobile the same minute. |
| **Account portability** | Cart, orders, wallet, loyalty, support tickets — all live in the shared backend, so a customer can start checkout on web and finish on mobile. |
| **Engineering velocity** | A shared `@xelnova/api` package means we never hand-write a duplicate API call. Adding a new endpoint is one place, not two. |
| **Future apps are cheap** | A future seller mobile app or business mobile app reuses the auth, navigation, query, and component patterns established here. |

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## Technology choices and rationale

| CONCERN | CHOICE | RATIONALE |
|---|---|---|
| Framework | Expo SDK 54 + React Native 0.81 | Single codebase for iOS + Android · Cloud builds via EAS · OTA updates without store reviews |
| Navigation | Expo Router (file-based, typed) | Compile-time route safety · Deep-linking out of the box · Mirrors Next.js conventions developers already know |
| Styling | NativeWind v4 | Same Tailwind classes as web — single design system, single mental model |
| Server state | TanStack React Query | Caching, optimistic updates, retries, infinite scrolling — production-grade defaults |
| Lists | Shopify FlashList | 60 fps scrolling on huge product grids; significant memory savings vs FlatList |
| Payments | Razorpay (WebView) | Reuses existing web checkout integration · PCI-DSS scope unchanged · No native SDK to maintain |
| Push | Expo Notifications + FCM/APNs | Single API, both platforms · Token registration handled in one place |
| Storage | `expo-secure-store` (tokens), `AsyncStorage` (preferences) | Tokens encrypted at rest in iOS Keychain / Android Keystore |
| Image upload | `expo-image-picker` + multipart | Reused by returns and ticket replies |
| Builds | Expo EAS Build & Submit | Cloud builds — no Mac required for iOS · Direct submit to both stores |

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## 3 · Phase-by-Phase Delivery

The work was split into four sequential phases, each ending in a typecheck-clean and lint-clean intermediate state. Every phase shipped end-to-end functionality — there is no scaffolding waiting for "later".

### Phase 0 · Foundation

| DELIVERABLE | DETAIL |
|---|---|
| Theme tokens | Brand palette, type scale, spacing extracted from web `globals.css` |
| NativeWind setup | Shared Tailwind preset across `mobile-web` and `ui-native` |
| Provider stack | SafeArea · gesture handler · React Query · auth · error boundary · onboarding gate · notifications bootstrap |
| `@xelnova/ui-native` v0 | 18 reusable components: Button · Card · Pill · Input · Skeleton · Avatar · Divider · RadioCard · Stars · RatingSummary · Image · Price · RatingPill · ProductCard · CategoryTile · EmptyState · SectionHeader · TagChip |
| Auth wiring | `@xelnova/api` configured with `expo-secure-store` token persistence; rehydrates session on launch |

### Phase 1 · Authentication

| SCREEN | CAPABILITY |
|---|---|
| Login | Email + password sign-in |
| Register | New customer onboarding with optional phone |
| Forgot password | Email-based reset trigger |
| Reset password | Token-based password set |
| Auth context | Persistent session via secure storage; auto-rehydrates on app open; logs out cleanly across all queries |

### Phase 2 · Browse & Discovery

| SCREEN | CAPABILITY |
|---|---|
| Home | Hero carousel · category strip · flash deals · featured · trending · new arrivals · best sellers · recently viewed (local) |
| Search | Debounced autocomplete · popular searches · recent searches with one-tap clear · filter sheet |
| Filter sheet | Sort · price range · rating · brand · category |
| Product detail | Image gallery · ratings & reviews · stock awareness · seller card · highlights · related products · share |
| Store detail | About · categories · featured products · full catalog · navigation tabs |
| Categories | Top-level grid + per-category browse with subcategories |

## Phase 3 · Cart, Checkout & Orders

| SCREEN | CAPABILITY |
|---|---|
| Cart | Live quantity sync · savings line · shipping config · proceed-to-checkout |
| Checkout · address | Saved-address picker with `RadioCard`s · "add new" inline |
| Checkout · review | Order summary · coupon application · payment method (COD or Razorpay) |
| Razorpay WebView | Hosted checkout in-app · bridge messages parsed for success / failure / dismiss |
| Checkout · success | Order confirmation · ETA · view-order CTA |
| Orders list | Status pills · image stacks · ship-by dates |
| Order detail | Timeline · shipment tracker · price breakdown · address · payment · retry-payment · invoice download · cancel · rate item · return / replace |

## Phase 4 · Account, Post-Purchase & Engagement

| SCREEN | CAPABILITY |
|---|---|
| Account home | 5-section menu · live loyalty pill · avatar · settings shortcut |
| Profile | Edit name, email, phone |
| Saved addresses | List · add · edit · delete · set default · **India-Post pincode auto-fill** |
| Wallet | Balance · paginated transactions · Razorpay top-up with KYC gate |
| Loyalty & referral | Balance · redeem-to-wallet · paginated ledger · referral code (copy + share) · apply code |
| Returns | List · create return with photo upload · reason selector |
| Reviews | Submit per-product reviews · PDP shows summary + approved reviews |
| Notifications | Paginated inbox · mark-as-read · deep-link routing · push registration |
| Support · chatbot | Conversational entry · auto-creates ticket on escalation |
| Support · list | Tickets with status pills |
| Support · detail | Threaded messages with role-aware bubbles · photo attachments · multi-photo reply |
| Support · manual | Direct ticket creation for users who skip chat |
| Account security | Change/set password · provider-aware (email vs Google vs phone) |
| Settings | Push toggle · language placeholder · terms / privacy / contact links · version · sign out · delete-account funnel |
| Onboarding | First-launch 3-slide intro · persisted skip flag |

## Cross-cutting capabilities

| CAPABILITY | IMPLEMENTATION |
|---|---|
| Push notifications | Auto-registers Expo Push Token after login; routes ORDER_*, RETURN_*, WALLET_*, TICKET_*, PROMO/OFFER notification types to the correct screen on tap |
| Deep linking | `xelnova://` scheme + universal links to `https://xelnova.in/*` — order/product/category links open in-app |
| Recently viewed | Last 20 products tracked locally; surfaced as a home rail |
| Recent searches | Last 10 queries surfaced on the search empty state |
| Sharing | Native share sheet for products and referral codes |
| Image upload | `expo-image-picker` → multipart → Cloudinary; reused by returns & ticket replies |
| Pagination | `useInfiniteQuery` for notifications, wallet, loyalty ledger |
| Error boundary | App-level catch with friendly retry UI |

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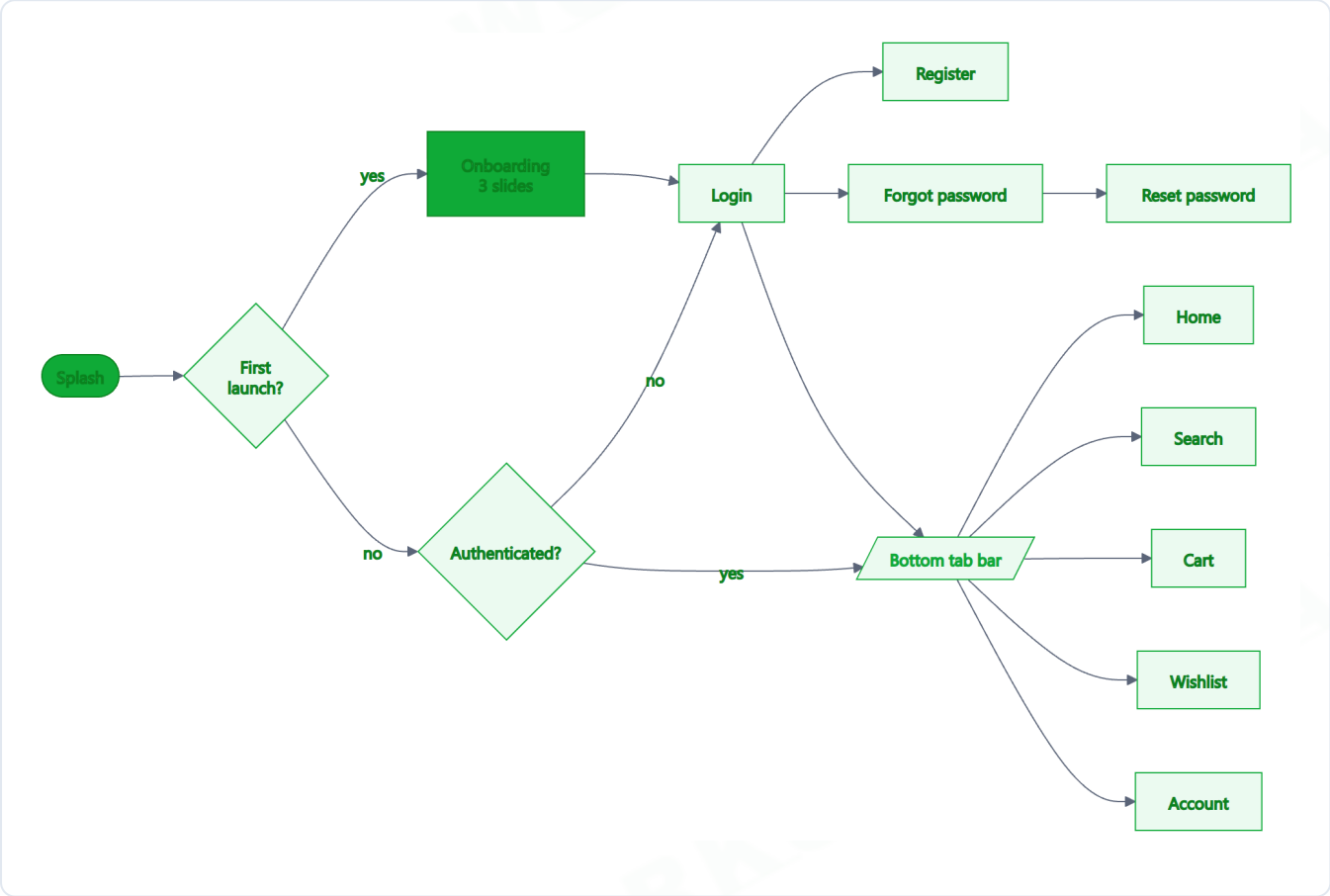
# 4 · Screen Flows

The following diagrams show the full information architecture of the app. Every box is a real screen that exists in the codebase; every arrow is a real navigation edge.

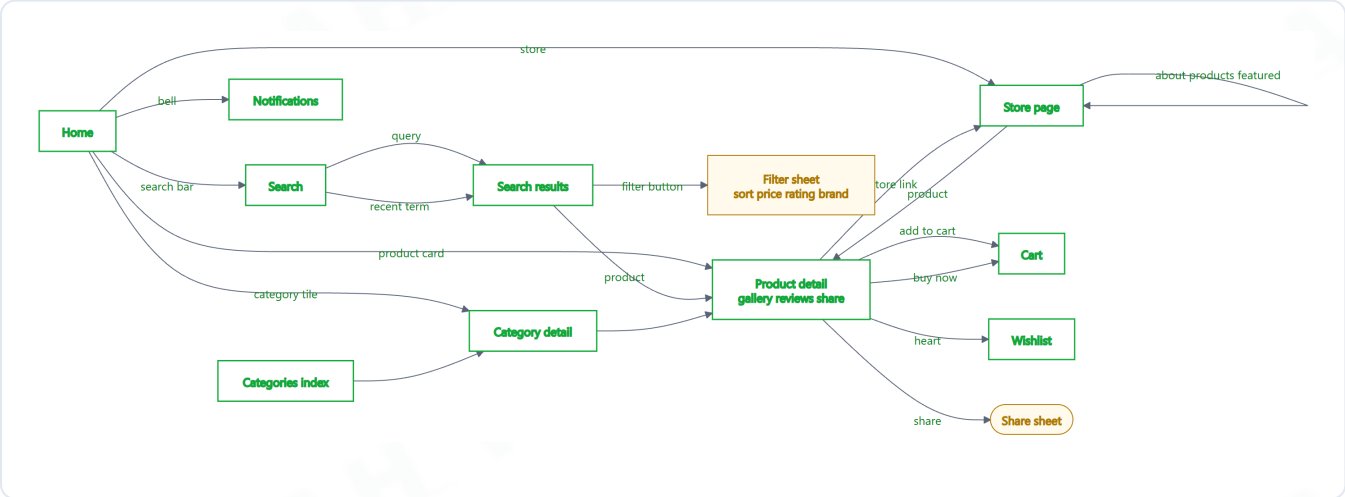
## 4.1 · Top-level app map

From cold-start splash through onboarding and authentication, into the five-tab bottom bar.



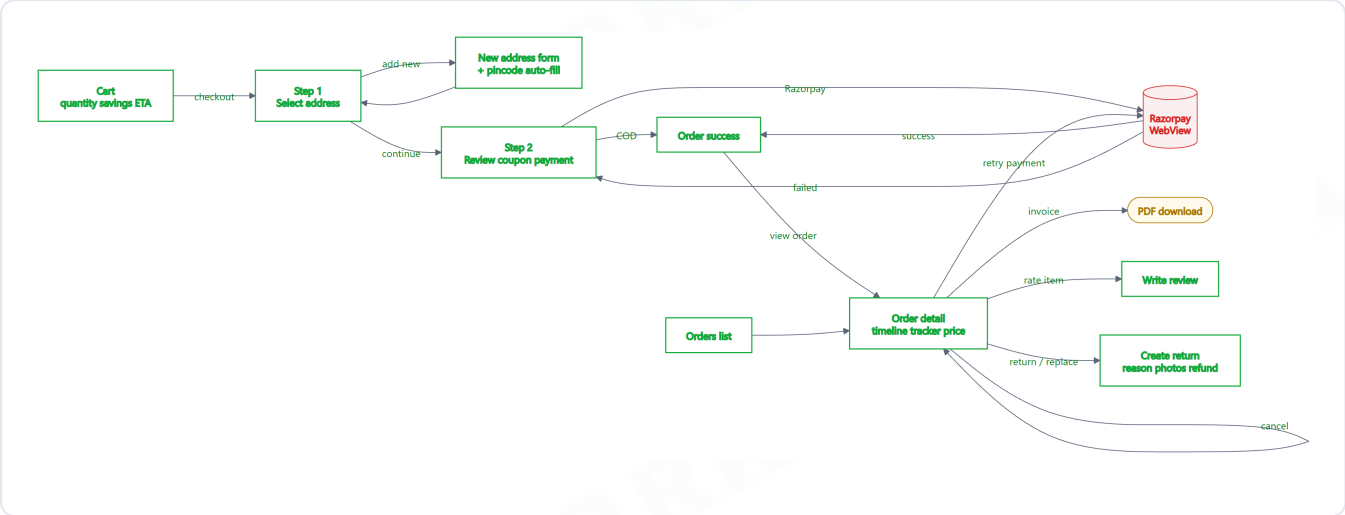

## 4.2 · Browse & discovery

How customers find products: home, search, filters, categories, store pages and the product detail page.



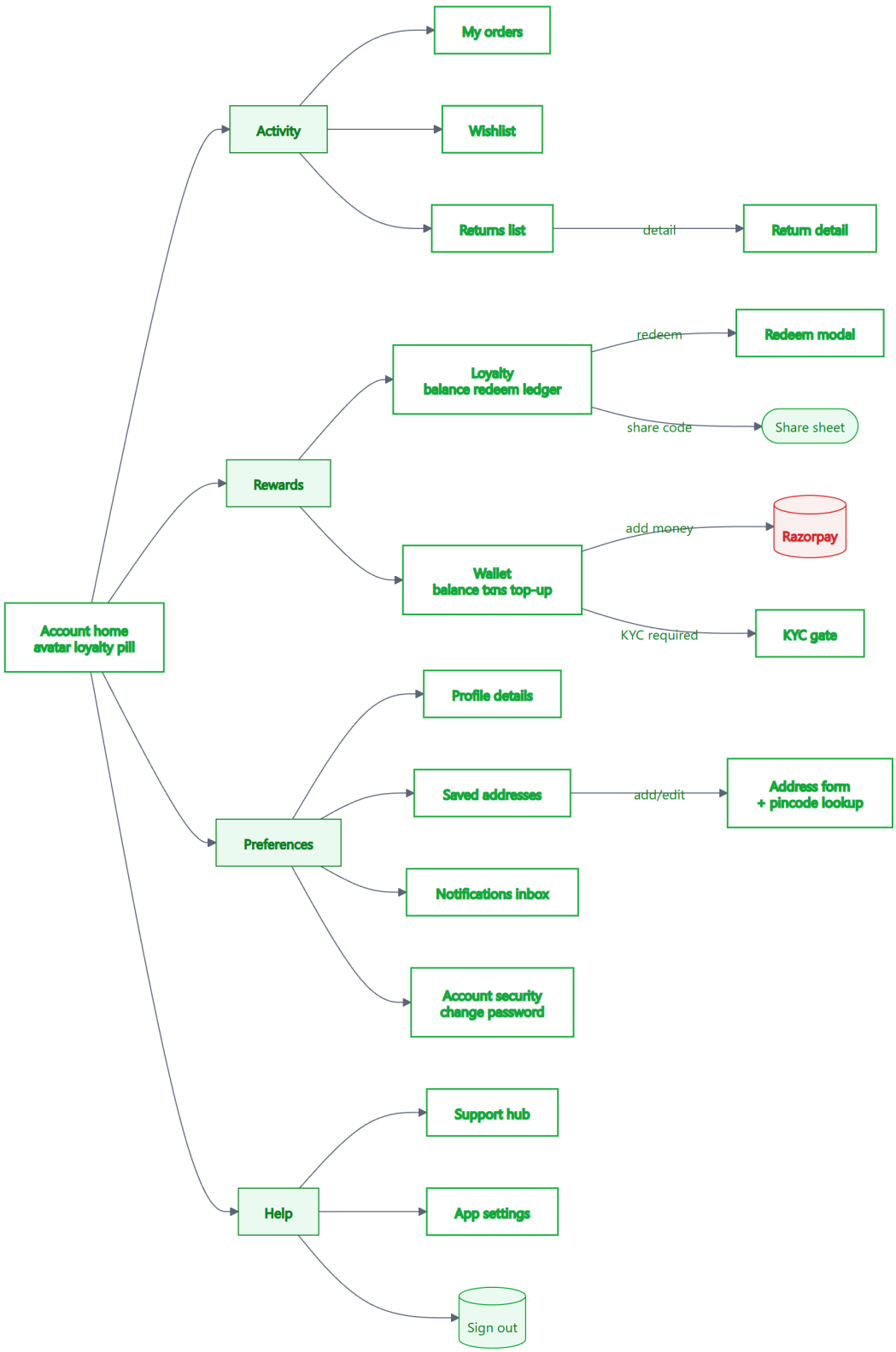
## 4.3 · Cart, checkout & orders

The end-to-end purchase journey including the Razorpay WebView, success state, and post-purchase actions on the order detail page.



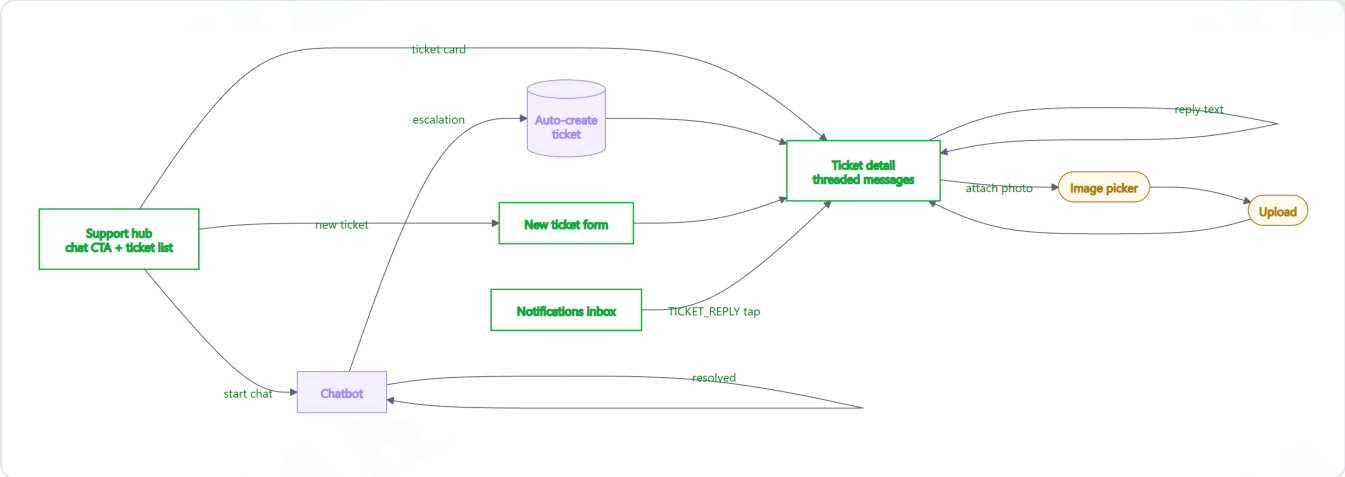

## 4.4 · Account hub

The account section, organised into Activity, Rewards, Preferences and Help groups — each fanning out to the corresponding screens.


Account home
avatar loyalty pill
Activity
My orders
Wishlist
Returns list
detail
Return detail
Rewards
Loyalty
balance redeem ledger
redeem
Redeem modal
share code
Share sheet
Wallet
balance txns top-up
add money
Razorpay
KYC required
KYC gate
Preferences
Profile details
Saved addresses
add/edit
Address form
+ pincode lookup
Notifications inbox
Account security
change password
Help
Support hub
App settings
Sign out

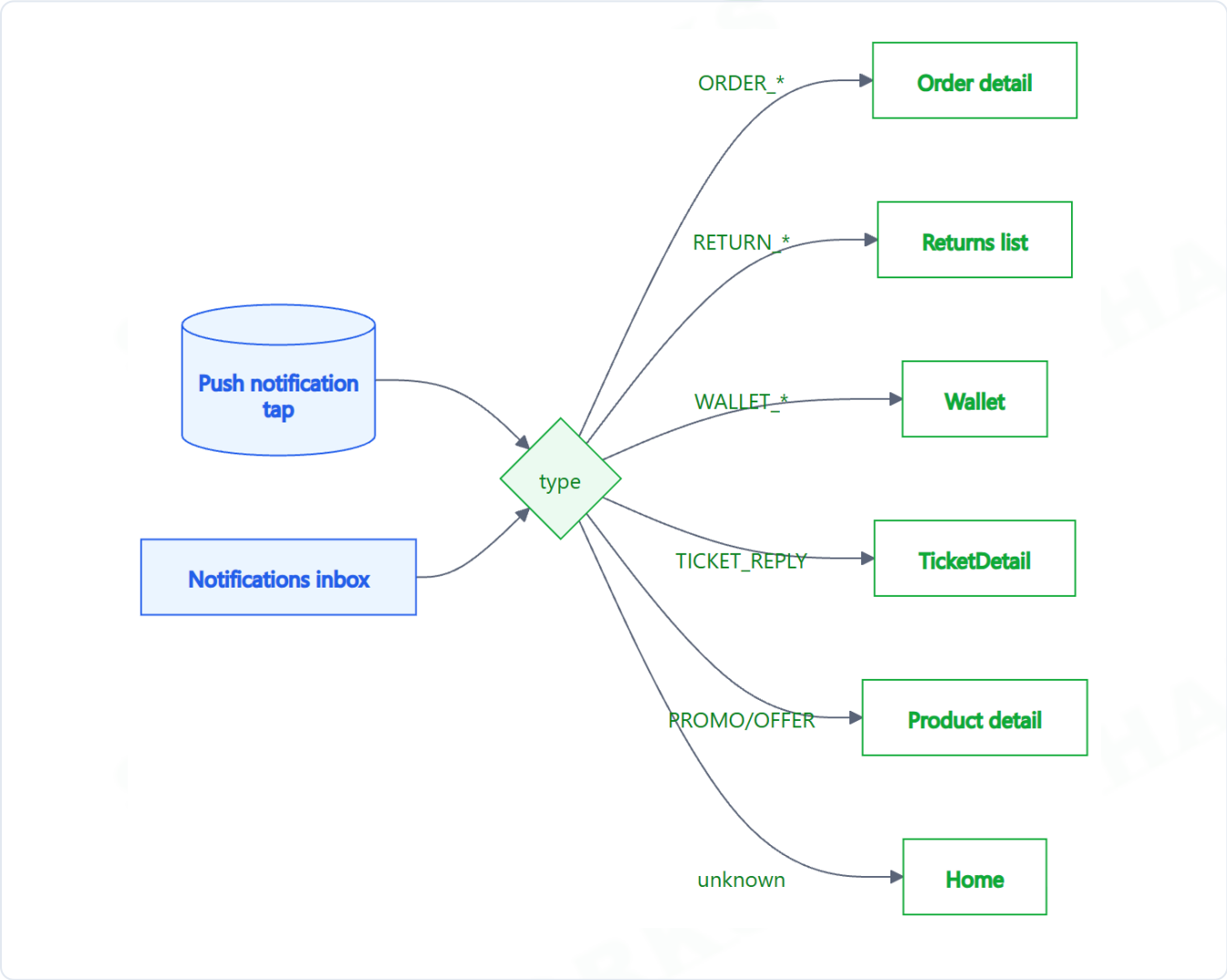
## 4.5 · Support flow

The support journey starts with the chatbot. Unresolved queries auto-create a ticket and the user is dropped into the threaded ticket detail screen.




## 4.6 · Push notification routing

How a tap on a push notification — or a tap on an in-app inbox row — gets routed to the correct screen based on the notification type.



# 5 · Quality & Engineering Discipline

## Type safety

| | |
|---|---|
| TypeScript strict mode | Enabled across `apps/mobile-web` and `packages/ui-native` |
| `tsc --noEmit` | Clean — 0 errors |
| Linter | Clean — 0 warnings |
| Expo Router typed routes | All 40+ routes type-checked at compile time — invalid `router.push()` calls fail the build |
| API contracts | 100 % reuse of `@xelnova/api` — no duplicate request code, no contract drift |

## Security

| | |
|---|---|
| Token storage | iOS Keychain · Android Keystore via `expo-secure-store` — never in plaintext |
| HTTPS-only | Backend pinned to `https://api.xelnova.in` |
| Razorpay | Hosted WebView checkout — card details never touch the app's process |
| Permissions | Photo, camera, notifications — all gated behind explicit user prompts with rationale strings |
| KYC | Wallet top-up gated by Aadhaar verification status |

## Performance

| | |
|---|---|
| List rendering | FlashList for product grids — keeps frame rate at 60 fps even on 1000-item lists |
| Image caching | `expo-image` with disk-backed cache — products cached after first view |
| Network resilience | React Query auto-retry on transient failures · per-screen error states · refresh-on-focus |
| Bundle size | Single JavaScript bundle, code-splitting handled by Expo for OTA updates |

## Resilience

| | |
|---|---|
| Error boundary | App-level catch with friendly retry UI; no red-screening in production |
| Pagination | Infinite-query patterns with end-reached and "load more" affordances |
| Stale-while-revalidate | Data shows immediately from cache, refreshes silently in the background |
| Offline tolerance | Cached data visible offline · graceful "no connection" states on critical screens |

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# 6 · Path to Launch

A realistic six-week launch plan from the current state to the public-facing app.

| | |
|---|---|
| WEEK 1 | **Build pipeline & assets** · `eas.json` profiles (development · preview · production) · final 1024 × 1024 icon · adaptive Android icon · splash artwork · APNs key + FCM `google-services.json` uploaded to EAS Credentials |
| WEEK 2 | **Internal alpha** · 10 stakeholders on TestFlight + Play Internal · bug bash · copy review · store-listing copy authored |
| WEEK 3 | **Closed beta** · 50 – 100 invited users · feedback collected via in-app support tickets · weekly bug-fix release |
| WEEK 4 | **Closed beta continues** · iterate on feedback · finalise screenshots · privacy policy and support URLs verified live |
| WEEK 5 | **Submit to stores** · App Store review (typically 24 – 72 hours) · Play Store review (typically same-day) · prepare release notes |
| WEEK 6 | **Public launch** · day-one analytics dashboard · push-notification announcement to existing web customers · monitoring alerts wired to Slack |

## Outstanding pre-launch items

The codebase is shippable. The remaining work is operational, not engineering.

| WORKSTREAM | OWNER | EFFORT |
|---|---|---|
| `eas.json` build profiles + `submit` config | Engineering | ~½ day |
| App Store Connect + Play Console listings | Marketing | 1 – 2 days |
| Final visual assets (icon, splash, screenshots) | Design | 2 – 3 days |
| APNs key + FCM credentials uploaded to EAS | Engineering | ½ day |
| Privacy policy + support URLs published | Legal/Marketing | 1 day |
| First production build + smoke test | Engineering | ½ day |

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# 7 · Roadmap Beyond MVP

What's in scope for the launch is the customer-MVP feature set — feature parity with web plus mobile-only enhancements. The following is a deliberately scoped *post-launch* roadmap.

## Quarter 1 (post-launch · stabilise & measure)

- **In-app analytics** — PostHog or Amplitude wired to all key events (add-to-cart, checkout, payment success, push opens)
- **Crash reporting** — Sentry for production crash visibility and diagnostics
- **Offline indicator** — `@react-native-community/netinfo` banner when the device is offline
- **Biometric quick re-login** — Face ID / fingerprint with `expo-local-authentication`
- **A/B test framework** — server-controlled variants for home rail order, hero copy, CTA wording

## Quarter 2 (engagement & differentiation)

- **Localisation** — Hindi · Marathi · Tamil · Bengali via `expo-localization`
- **In-app KYC** — Aadhaar verification via Digilocker WebView (today's flow uses the website)
- **Product comparison** — side-by-side compare of up to 3 products
- **Wishlist folders & sharing** — organise saved items, share lists with friends
- **Stories / video commerce** — vertical video PDP enrichment for fashion and lifestyle SKUs

## Quarter 3 (scale & efficiency)

- **OTA update channel** — staged rollouts for hot-fixes without going through store review
- **Background fetch** — pre-warm cart and orders before the user opens the app
- **Personalised home** — server-side ML-driven product rails based on view + purchase history
- **Razorpay native SDK** — replace WebView with native module for ~1 second faster checkout

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# 8 · Appendix · Stack & Dependencies

## Runtime dependencies (production)

| PACKAGE | VERSION | PURPOSE |
|---|---|---|
| `expo` | ~54.0 | App framework |
| `react-native` | 0.81 | Native runtime |
| `expo-router` | ~6.0 | File-based navigation |
| `nativewind` | ^4.2 | Styling |
| `@tanstack/react-query` | ^5.59 | Server state |
| `@shopify/flash-list` | 2.0 | High-performance lists |
| `react-native-webview` | 13.15 | Razorpay + KYC hosted flows |
| `expo-secure-store` | ~15.0 | Encrypted token storage |
| `@react-native-async-storage/async-storage` | 2.2 | Local preferences |
| `expo-notifications` | ~0.32 | Push delivery |
| `expo-image-picker` | ~17.0 | Camera & gallery |
| `expo-image` | ~3.0 | Cached image rendering |
| `expo-clipboard` | latest | Copy referral codes |
| `lucide-react-native` | ^1.16 | Icon library |
| `@xelnova/api` | workspace | Shared HTTP client |
| `@xelnova/ui-native` | workspace | Shared component library |

## Backend integration points

The mobile app talks to **47 distinct REST endpoints** across these modules — every one of them was already in production for the web app:

`auth` · `products` · `categories` · `cart` · `orders` · `users` · `search` · `reviews` · `payment` · `returns` · `wallet` · `tickets` · `notifications` · `wishlist` · `verification` · `stores` · `upload`

## Repository hygiene

- All new code is **TypeScript strict** and lint-clean
- File naming follows the existing monorepo conventions (kebab-case for files, PascalCase for components)
- Every screen has a corresponding entry in `.expo/types/router.d.ts` — invalid `router.push()` calls are caught at compile time
- Every shared component lives in `packages/ui-native` and is unit-style typed against React Native's `ViewProps` / `TextProps` /etc.

**End of document**

Xelnova Customer Mobile App · Detailed Roadmap · v1.0

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